Jinpeng Lu

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Multimodal Foundation Models | Generative World Models | Embodied AI

Research Profile & Core Strengths

My research centers on **multimodal foundation models**, **generative world models**, and **embodied AI**, with an emphasis on reliable representations for dynamic environments, video understanding/generation, and model evaluation.

My work converts open-ended capability questions into reproducible benchmarks, diagnostic evidence chains, and efficient model designs, including WRBench dynamic world-state evaluation, VeloxSeg, DINOv3 medical transfer analysis, and H2ASeg PET/CT tumor segmentation.

Experienced with PyTorch / DDP / CUDA / Linux training and evaluation pipelines, data construction, custom model implementation, ablation analysis, visualization, and English academic manuscript writing.

Education

University of Science and Technology of China — Sep. 2025 – Present
M.S. in Information and Communication Engineering, expected Jun. 2028 — *Advisor: Zhiwei Xiong*

Harbin Institute of Technology, Shenzhen — Sep. 2021 – Jun. 2025
B.S. in Data Science and Big Data Technology; GPA: 3.728/4, Rank: 17/172, Top 10% — *Mathematics / Statistics / Numerical Computing*

Representative Works

Generative World Models

Current World Models Lack a Persistent State Core, *First Author* — **arXiv 2026**
- **Task:** Evaluate whether generative video models maintain off-screen dynamic world states under viewpoint changes, beyond locally plausible videos.
- **Method:** Built **WRBench** by treating camera motion as an **observability intervention** and defining a six-dimensional diagnostic profile for re-observation consistency.
- **Result:** Evaluated **23 models** across **four control paradigms** and **9,600 videos**, calibrated with **2,547 human verdicts**, identifying endpoint-binding failures in current models.

Medical Vision Foundation Models

VeloxSeg: JL-Guided Efficient 3D Medical Segmentation, *First Author* — **ICLR 2026**
- **Task:** Improve the efficiency-robustness tradeoff in high-dimensional multimodal 3D perception, including PET/CT and MRI segmentation.
- **Method:** Designed a dual-stream CNN-Transformer framework with **Paired Window Attention**, **JL-guided convolution**, and **SDKT**.
- **Result:** Achieved a **26% Dice gain** with **1.66M parameters** and improved GPU / CPU throughput by **11x / 48x**.

Does DINOv3 Set a New Medical Vision Standard?, *Co-first Author* — **arXiv 2025**
- **Task:** Evaluate whether DINOv3 can serve as a general medical vision foundation model across heterogeneous 2D/3D medical tasks.
- **Method:** Co-built a transfer benchmark covering **classification, segmentation, and registration** across X-ray, CT, endoscopy, WSI, EM, PET, and ultrasound.
- **Result:** Found partial transfer of 2D self-supervised features to 3D tasks, alongside domain shift and non-monotonic scaling behavior.

H2ASeg: Hierarchical PET/CT Tumor Segmentation, *First Author* — **MICCAI 2024**
- **Task:** Improve PET/CT tumor segmentation where PET provides functional cues and CT provides anatomical structure, but modality fusion is unstable.
- **Method:** Designed hierarchical cross-modal interaction with **MCSA/BDSA** and **Target-Aware Modality Weighting**.
- **Result:** Achieved a **20.25% absolute Dice improvement** on AutoPET-II, with module ablations yielding **12.20% / 10.94% Dice gains**.

Core Methods & Engineering

Research: World-model evaluation, VLM/VLA representation diagnostics, medical vision foundation-model transfer, PET/CT and MRI 3D segmentation

Engineering: Python, PyTorch, CUDA, DDP, Shell/Bash, LaTeX; training/evaluation pipelines, experiment reproduction, ablation studies, custom model implementation

Tools: Linux, Git, Docker, Weights & Biases / TensorBoard, vision, multimodal, and medical-imaging data-processing toolchains

Writing: English manuscript writing for ICLR / MICCAI / Medical Image Analysis submissions, experiment analysis, and reproducibility documentation

Full Research Outputs

Generative World Models — 2026 – Present

[1] **Current World Models Lack a Persistent State Core**, *First Author* — **arXiv 2026**

Contribution Introduced WRBench for dynamic world-state evaluation under controlled viewpoint changes with a six-dimensional diagnostic profile.

VLM & VLA Representation Learning — 2026 – Present

[1] **VLA-Trace: Diagnosing Vision-Language-Action Models through Representation and Behavior Tracing**, *Co-author* — **arXiv 2026**

Contribution Contributed to connecting representation tracing with rollout behavior to diagnose routing, grounding, and semantic-following failures in VLA models.

[2] **Pelican-Unify 1.0: A Unified Embodied Intelligence Model**, *Co-author* — **arXiv 2026**

Contribution Contributed to unified embodied model training and evaluation across understanding, reasoning, future imagination, and action generation.

Medical Vision Foundation Models — 2023 – 2026

[1] **Johnson-Lindenstrauss Lemma Guided Network for Efficient 3D Medical Segmentation**, *First Author* — **ICLR 2026**

Contribution Proposed VeloxSeg for efficient multimodal 3D medical segmentation under a lightweight parameter budget.

[2] **Does DINOv3 Set a New Medical Vision Standard?**, *Co-first Author* — **arXiv 2025**

Contribution Introduced a DINOv3 transfer evaluation across medical modalities and identified where foundation-model scaling remains unreliable.

[3] **H2ASeg: Hierarchical Adaptive Interaction and Weighting Network for Tumor Segmentation in PET/CT Images**, *First Author* — **MICCAI 2024**

Contribution Proposed H2ASeg for PET/CT tumor segmentation through hierarchical multimodal interaction and tumor-aware weighting.

[4] **AttriMIL: Revisiting Attention-based Multiple Instance Learning for WSI Classification**, *Co-author* — **Medical Image Analysis 2025**

Contribution Contributed to revisiting attention-based multiple-instance learning for interpretable whole-slide image classification.

[5] **IPGPhormer: Interpretable Pathology Graph-Transformer for Survival Analysis**, *Co-author* — **arXiv 2025**

Contribution Contributed to interpretable pathology graph modeling for survival analysis and prognosis prediction.

[6] **A Localization-to-Segmentation Framework for Automatic Tumor Segmentation in Whole-Body PET/CT Images**, *Co-author* — **arXiv 2023**

Contribution Contributed to a localization-to-segmentation cascade for automatic whole-body PET/CT tumor segmentation.

Additional Project

Generative Sensor Signal Enhancement | *Lead, National Undergraduate Innovation Program* — Jun. 2023 – Jun. 2024

- Reproduced and compared GAN, diffusion, and flow-based models; designed conditional guidance strategies to improve reconstruction quality and perceptual fidelity, producing a reproducible comparison pipeline and experiment report.